Reliability of the Incremental Costs Effectiveness Ratio

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1. Calculate the event-proportions (p1 and p2) and the mean costs (m1 and m2) in both treatment groups and calculate the ICER as (m1-m2)/(p1-p2).

Group 1

Event proportion group 1 = 0.7476
Mean cost group 1 = €49786.65
95% confidence interval for mean cost group 1 = [€29115.4 - €70457.9]

Group 2

Event proportion group 2 = 0.6355
Mean cost group 2 = €24805.67
95% confidence interval for mean cost group 2 = [€23834.41 - €25776.92]

Looking at the differences between the groups, the new treatment seems more effective (75% survival vs 64% survival) but also more expensive (€49,8k vs €24,8k). The next step is calculating the ICER ratio

1. ICER ratio

If we look at the difference between the groups but now incremental, we see an incremental effectiveness of 0.1121 and an incremental cost of $24,980.99 . With these numbers we can calculate the ICER. This is the ratio that combines both incremental numbers into a ratio.

ICER = incremental cost / incremental effectiveness

ICER = 24980,99 / 0.1121 = €222,835.93

This ratio tells us that we have to spend €222,836.93 more to get one additional patient to survive the one year mark compared to the standard treatment.

2. Use theory of transformations and the delta-method to derive an asymptotic 95% confidence interval (list and discuss assumptions that are made).

Assumptions and why it matters:

1. Approximate normality as sample *n* becomes larger
   - Currently our statistic involves a non-linear transformation, so we use the Delta method to linearizes a nonlinear function around the asymptotic normal estimators. This matters because the Delta method uses a first-order Taylor expansion around those asymptotic normal estimators.
2. Both groups are statistically independent
   - This allows us to set all covariance terms to 0 and easily sum the variances of each group upon calculating the variance of the difference in means or proportions.
3. Differentiability of the ICER function
   - The ICER function must be differentiable at the true parameter values. Differentiability gives us a valid linear approximation, which the delta function relies on.

2. Use theory of transformations and the delta-method to derive an asymptotic 95% confidence interval (list and discuss assumptions that are made).

Assumptions and why it matters:

4. The effectiveness ratio is not close to zero
   - If delta effectiveness comes too close to 0, the ICER ratio becomes unstable and the CI can become extremely wide or entirely meaningless.
5. Costs have finite variances
   - The Delta method assumes that the the variables mean and costs have finite variances. If they were to have infinite variances the CI would become meaningless because the sampling variabilityof the mean would not decrease as the sample size increases.

2. Use theory of transformations and the delta-method to derive an asymptotic 95% confidence interval (list and discuss assumptions that are made).

We calculated the variances of the costs and proportions in both groups. After this, we calculated the covariances of both groups.

These numbers can be implemented to calculate the asymptotic 95% CI.

This asymptotic confidence interval happened to be [€ -31,011.38 - €476,683.24]. This confidence interval is so wide that we can conclude that our point estimate is not reliable and gives us little information.

The next step is the non-parametric method bootstrapping. We compute new estimates in order to compare these with a parametric distribution method later.

```
Variance of costs in group 1: 22913358223.36
Variance of costs in group 2: 49848296.96
Variance of proportions in group 1: 0.1887
Variance of proportions in group 2: 0.2316
Covariance (costs, events) in group 1: 3205.64
Covariance (costs, events) in group 2: -1873.91
Covariance (dC, dE): 6.3302
Delta Cost = 24980.99, Delta Effect = 0.1121
ICER = €222835.93
Asymptotic 95% CI (delta): €-31011.38 to €476683.24
```

3. Bootstrap sampling

We resampled the data with 5000 iterations and calculated the ICER for each case. Next, we calculated the 95% confidence interval from the distriubtion of bootstraps calculated ICERs. To do so, we used the percentile method.

Bootstrap 95% Confidence Interval for ICER (Percentile Method): [€48061.6, €1032199.33]

3. Bias check and correction

We performed a bias check by calculating the difference between the mean bootstrap estimates and our original estimates. We ended up with a bias of 0.48%, which is very small. So small that the presence is barely noticable and therefore we have made the final decision to ignore the bias.

```
Mean of bootstrap ICER estimates: €239581.27
Original sample ICER: €222835.93
Bias of bootstrap ICER estimates: €16745.34
Standard error of bootstrap ICER estimates: €3466378.92
Absolute bias relative to standard error: 0.48%

Bias correction does not appear to be necessary based on the relative bias
```

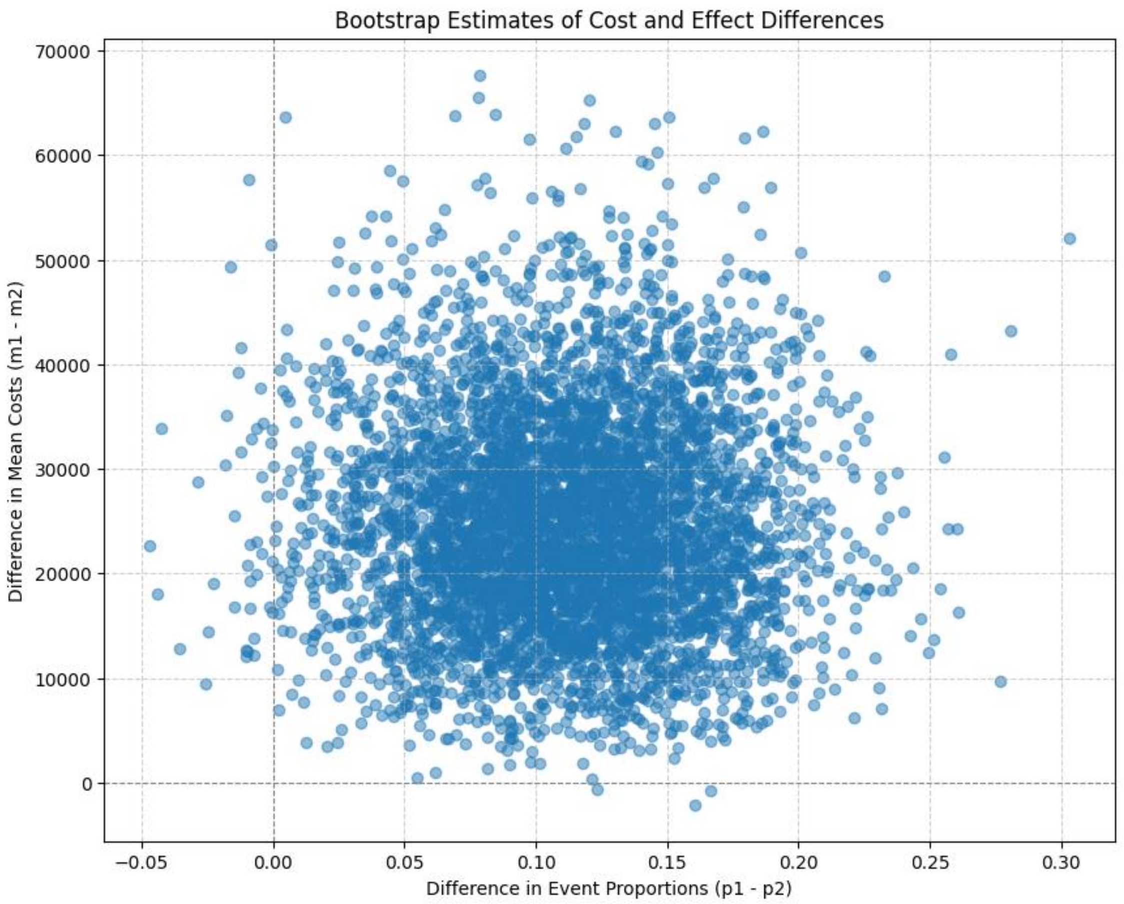
3. Plot bootstrap estimates

Almost all the blue dots are in the top-right quadrant of the plot. This means the incremental cost is positive and the incremental effect is positive.

Therefore, we can conclude that the treatment of group 1 is more costly but also more effective than the standard treatment.

However, we also see that the range of the cost of the treatment is still quite large and therefore the 'price' of the benefit is uncertain.

In the next step we will further investigate this plot with help of a quadrant analysis



3. Quadrant analysis

Using quadrant analysis we try to count the blue dots in each quadrant of the plot. The results are shown in the screenshot on the right.

As we can see, 99.1% is on Quadrant 1. Conforming what we already saw in the previous slide. This group is more costly but also more effective.

Additionally, 0.82% is in Quadrant 4. Here, the new treatment is more costly and less effective.
Only 0,06% landed in Quadrant 2. Suggesting the new treatment is less costly and more effective.

```
Quadrant Analysis of Bootstrap Estimates:
Quadrant 1 (More costly, More effective): 4956 (99.12%)
Quadrant 2 (Less costly, More effective - Dominant): 3 (0.06%)
Quadrant 3 (Less costly, Less effective): 0 (0.0%)
Quadrant 4 (More costly, Less effective - Dominated): 41 (0.82%)
On axes: 0 (0.0%)
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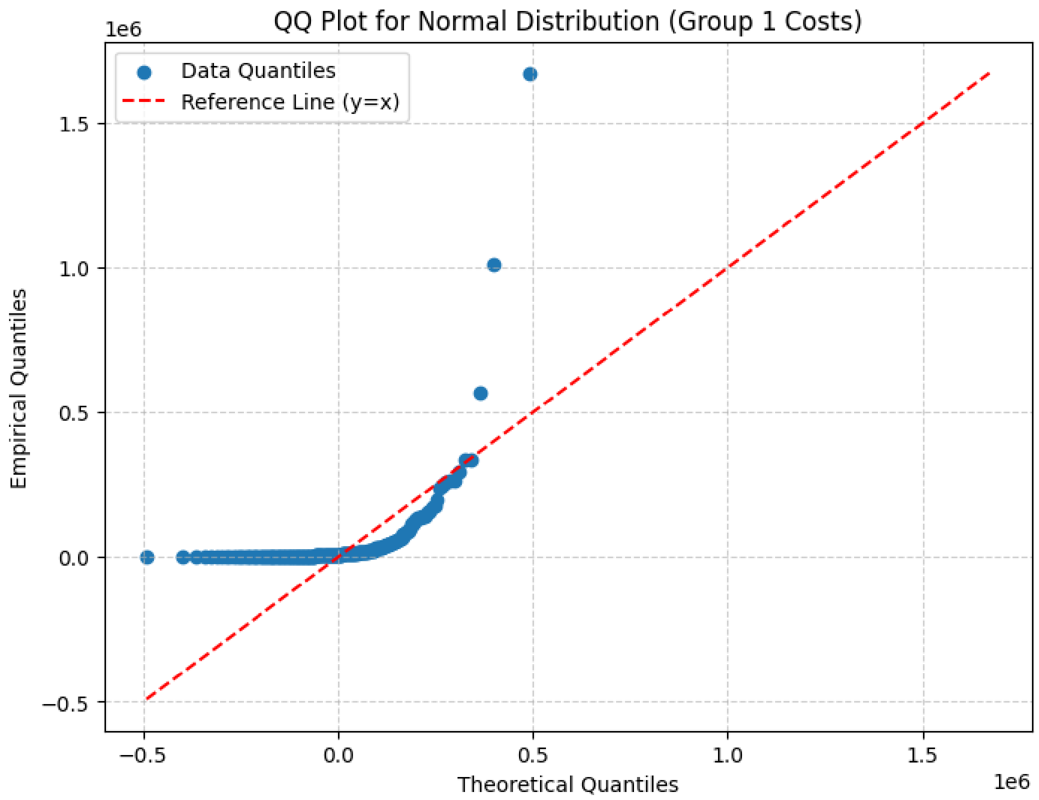
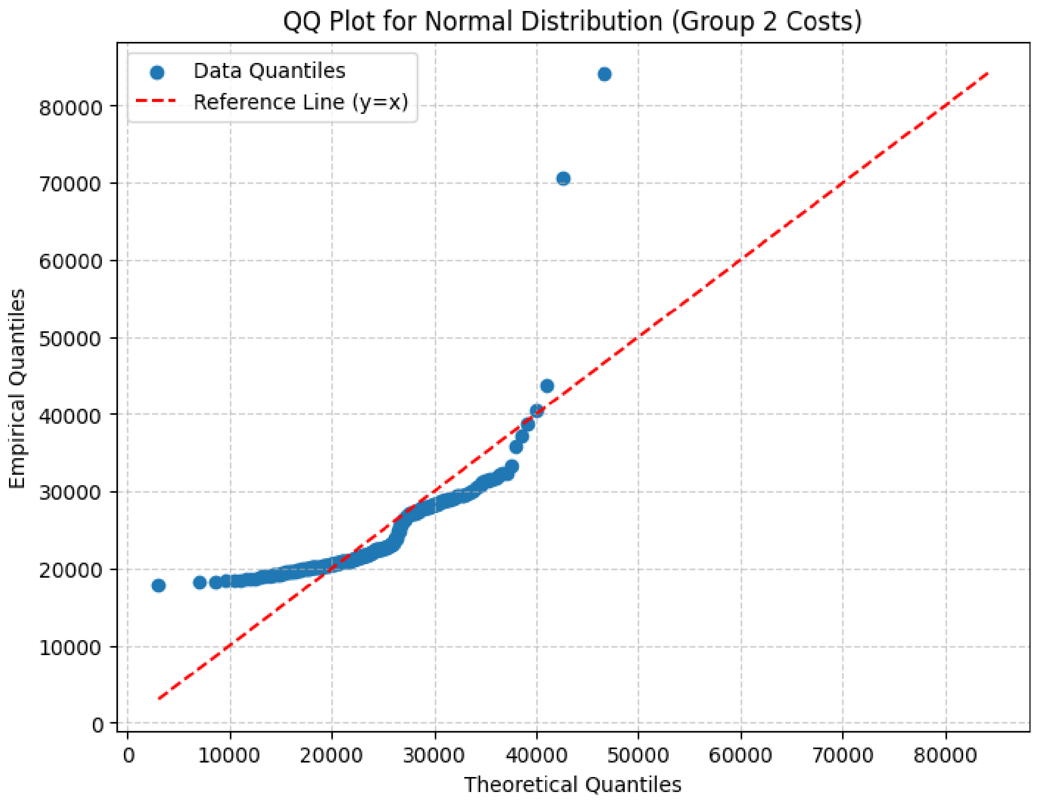
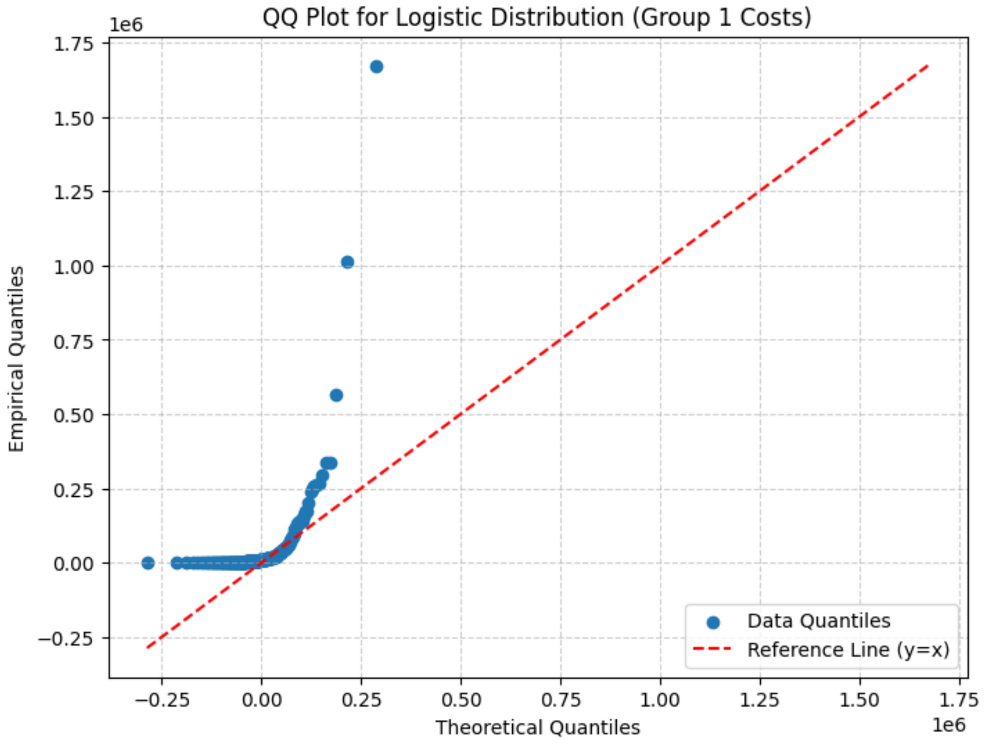
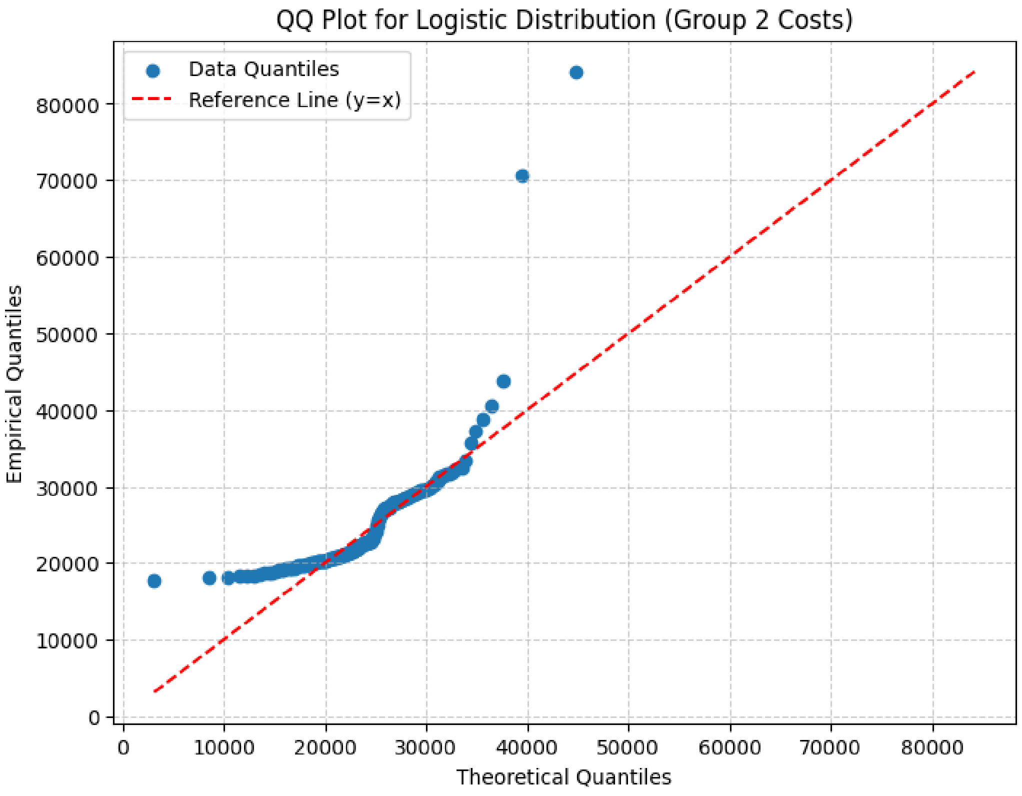
4. Parametric distributions - Normal



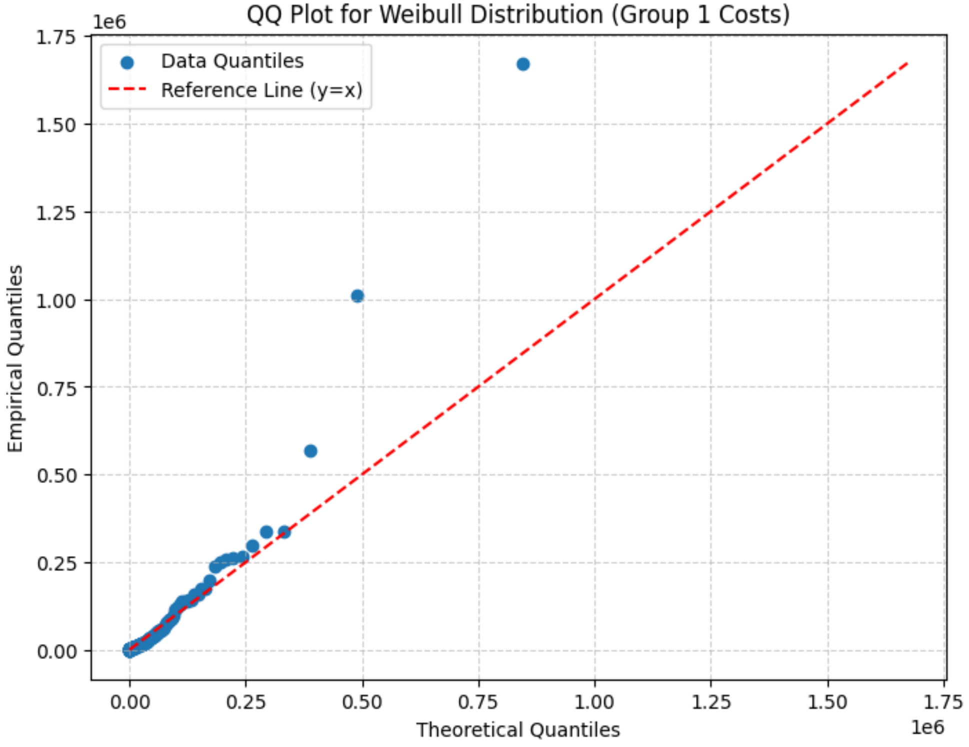


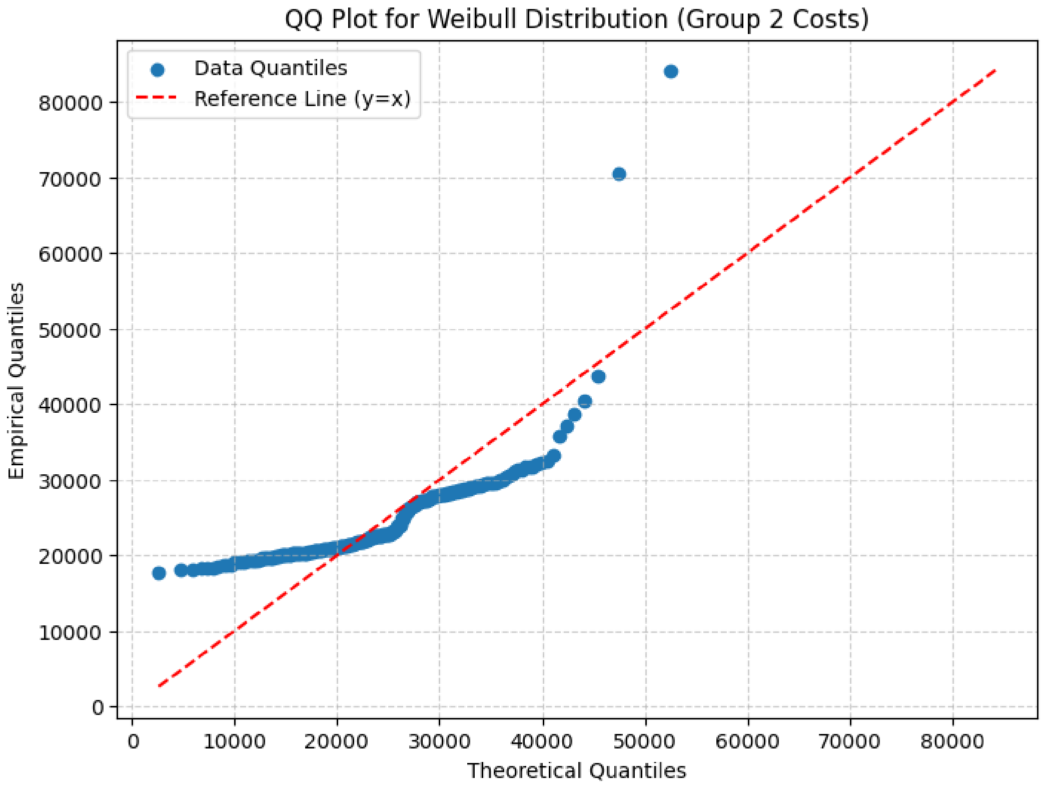
4. Parametric distributions - Logistic



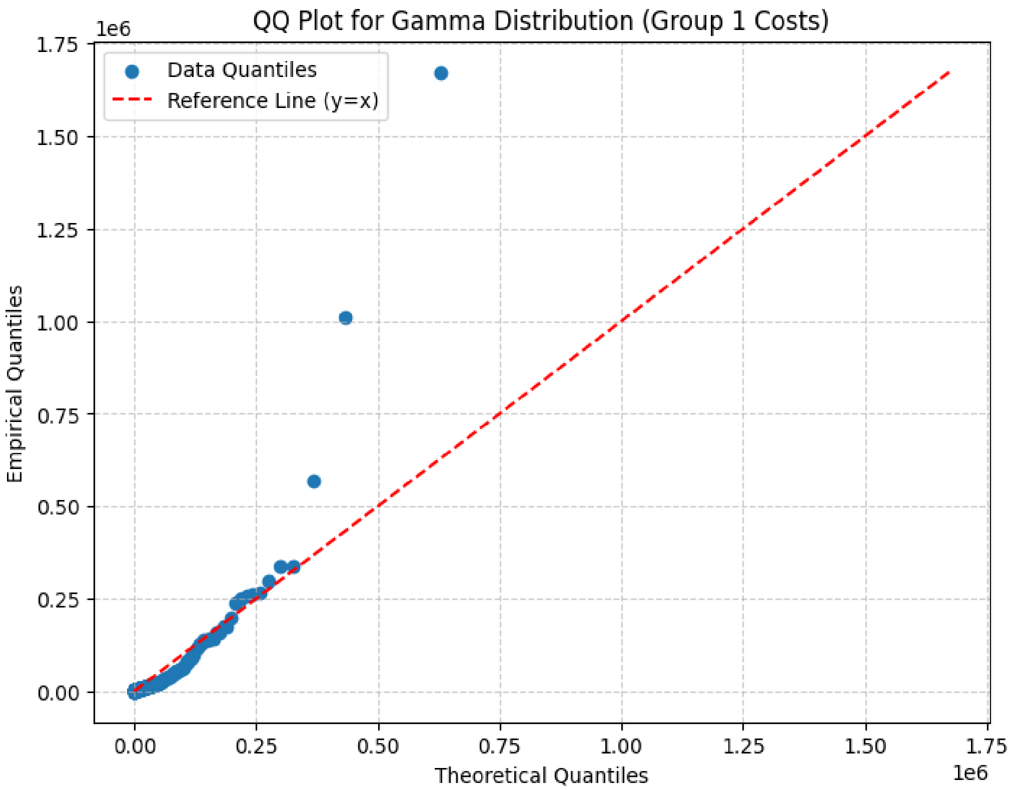


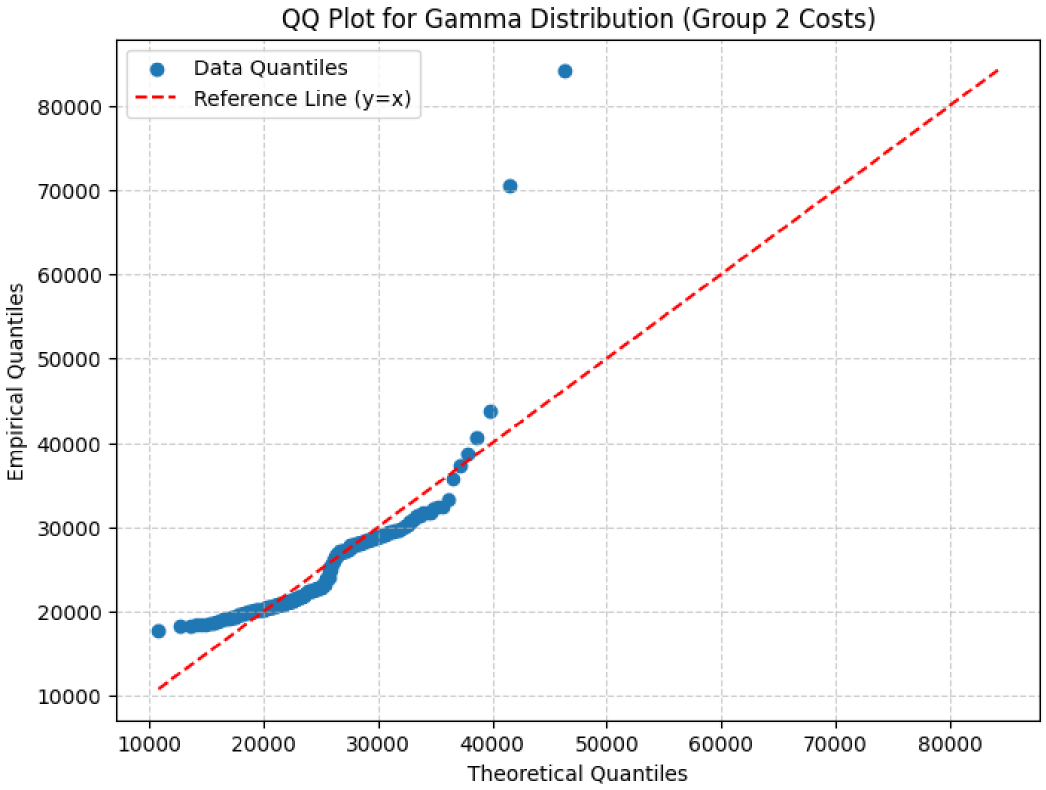
4. Parametric distributions - Weibull



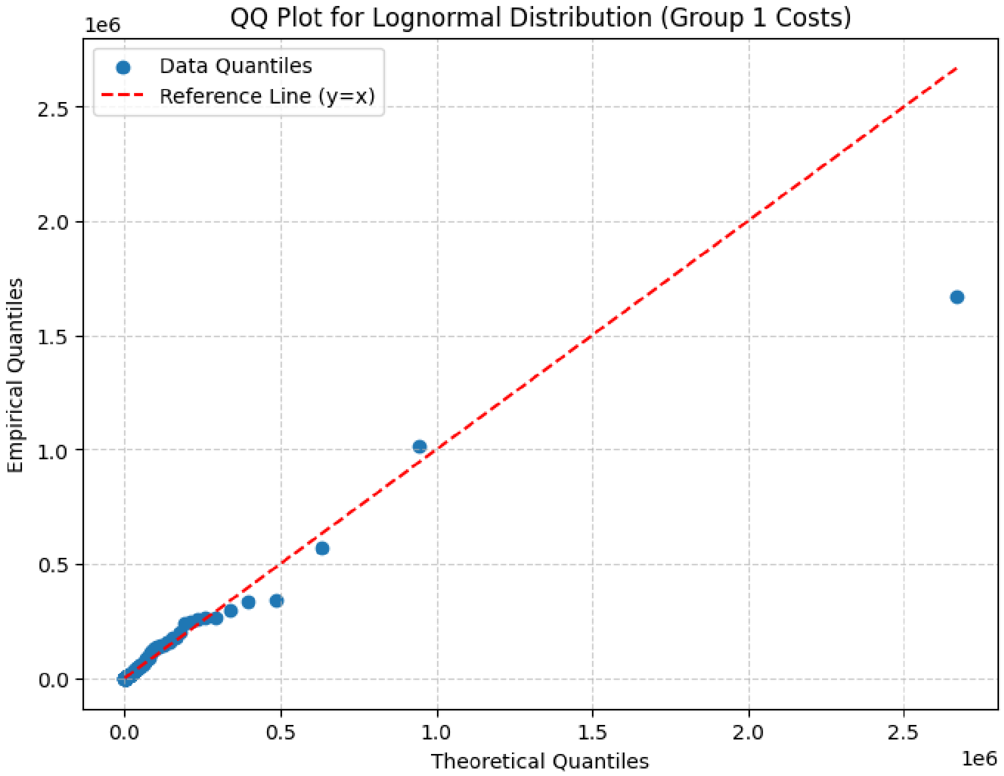


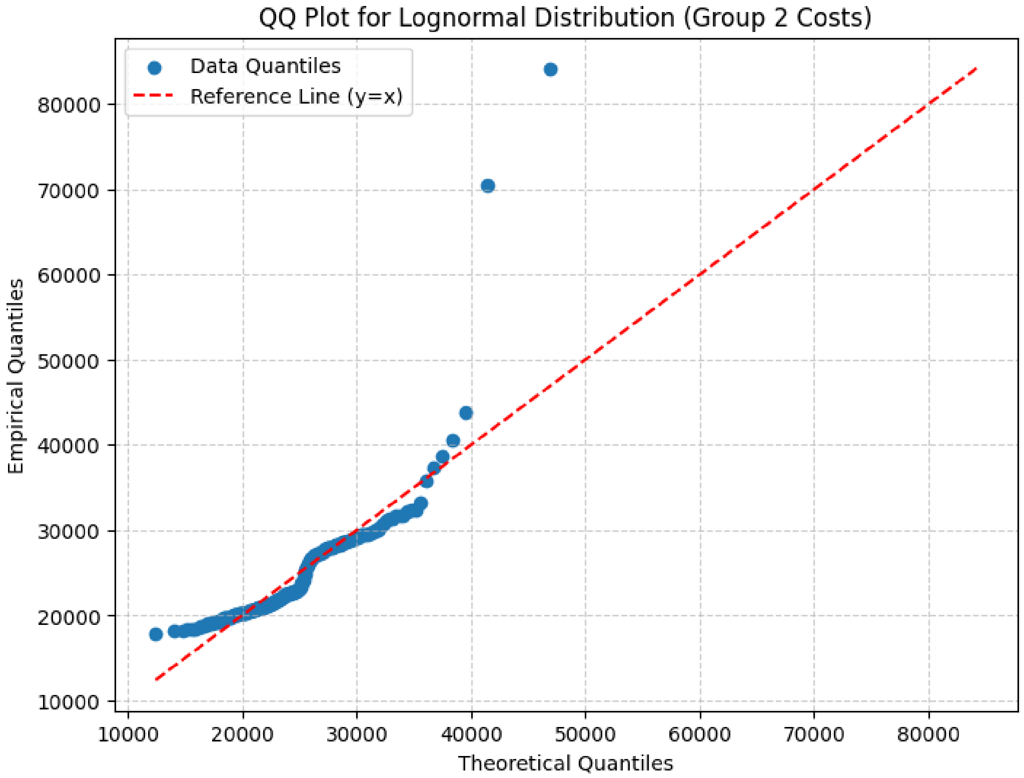
4. Parametric distributions - Gamma





4. Parametric distributions - Lognormal





4. Best model: lognorm and gamma

The visual inspection of the distributions displays the potential of the lognorm and gamma distributions, but also the poor performance of the normal, logistic and lognorm distributions.

The poorly fitting models showed clear deviations from the reference line, indicating poor fit due to skewness. The lognorm and gamma model follow the reference line more closely, indicating that these 2 are the best performing distributions for both groups.

Next step is simulation where we will draw both the Lognormal and Gamma model

5. Parametric simulation of 1000 times sampling

For each of the simulation iterations:

- The cost values for each group were drawn from the fitted parametric distributions
- Event outcomes were simulated using Binomial distributions with the event proportions
- The mean costs and event proportions were calculated per simulation, and the ICER was computed everytime as well. Simulations where the denominator (the effectiveness ratio) was near zero were excluded to avoid unstable ratios. This was not the case and we were able to retrieve 1000 valid ICER values for both the Lognormal and Gamma model.

5. Draw cost-data from estimated distributions

The cost data were summarized as:

Lognorm

Group 1

Mean = €50475.66

SD = €16263.20

95% CI = [30720.27, 89668.96]

Group 2

Mean = 24703.64

SD = 373.74

95% CI = [23983.55, 25439.61]

Gamma

Group 1

Mean = €49666.33

SD = €5536.86

95% CI = [39612.54, 61152.46]

Group 2

Mean = €24835.66

SD = €403.37

95% CI = [24118.96, 25690.54]

5. Draw events from binomial distributions with p1 and p2

The events proportions were summarized as

Lognorm

Group 1

Mean = 0.7454

SD = 0.0292

95% CI = [0.6892, 0.8010]

Group 2

Mean = 0.6369

SD = 0.0322

95% CI = [0.5714, 0.6947]

Gamma

Group 1

Mean = 0.7477

SD = 0.0296

95% CI = [0.6845, 0.8010]

Group 2

Mean = 0.6370

SD = 0.0339

95% CI = [0.5665, 0.6996]

5. Calculate ICERs for each simulation

The simulated ICER values were summarized as

Lognorm

Mean= €551053.24

SD = €10893950.61

95% CI = [47105.38, 987970.44]

Gamma

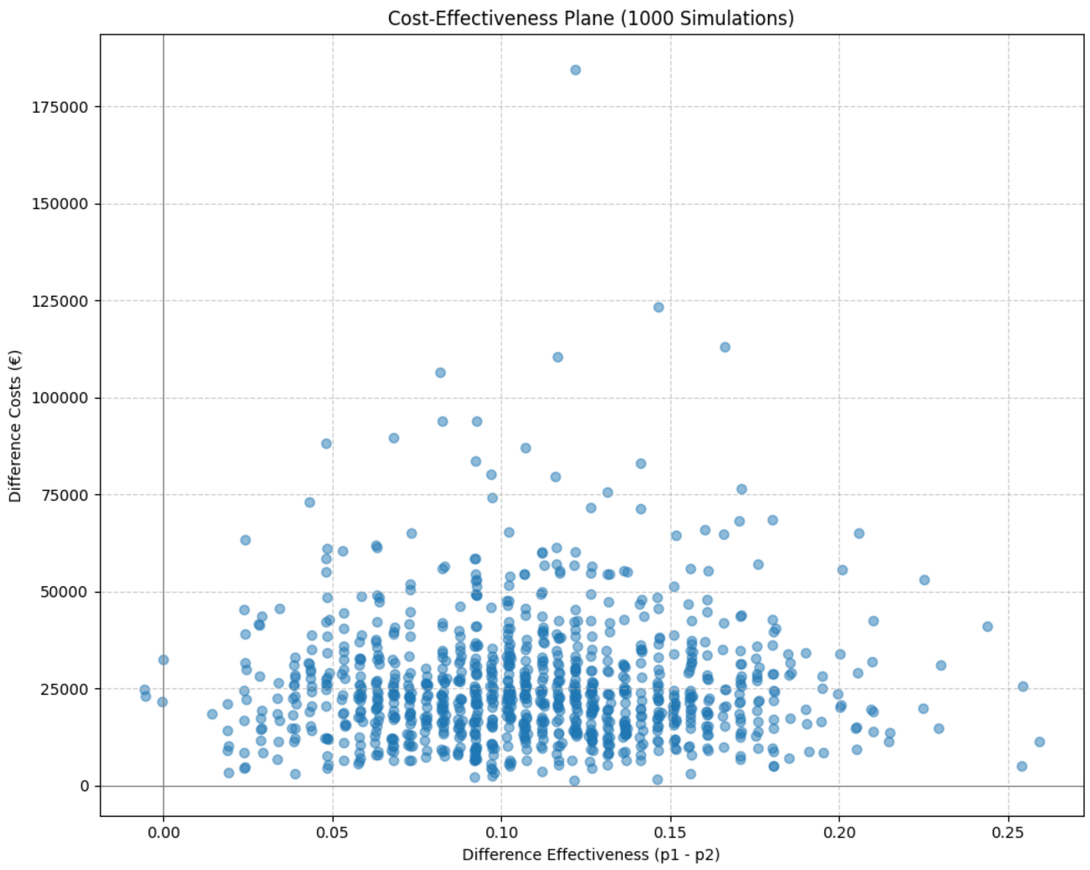
Mean = €37870.69

SD = €6358229.18

95% CI = [93913.69, 886374.49]

5. Comparison Bootstrap & Paramatric simulation

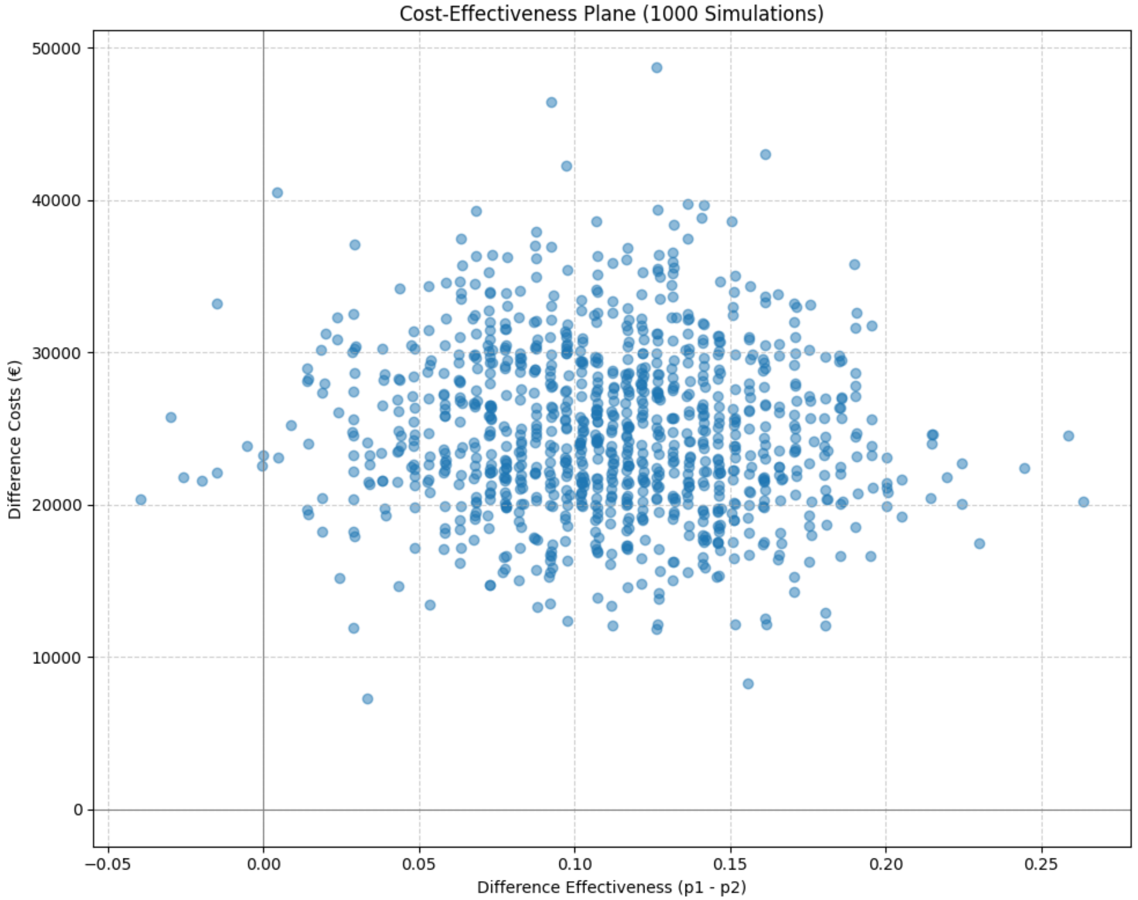
Lognorm




```
Quadrant Analysis of Simulated Estimates:
Quadrant 1 (More costly, More effective): 997 (99.7%)
Quadrant 2 (Less costly, More effective - Dominant): 0 (0.0%)
Quadrant 3 (Less costly, Less effective): 0 (0.0%)
Quadrant 4 (More costly, Less effective - Dominated): 3 (0.3%)
On axes: 0 (0.0%)
```

5. Comparison Bootstrap & Paramatric simulation

Gamma



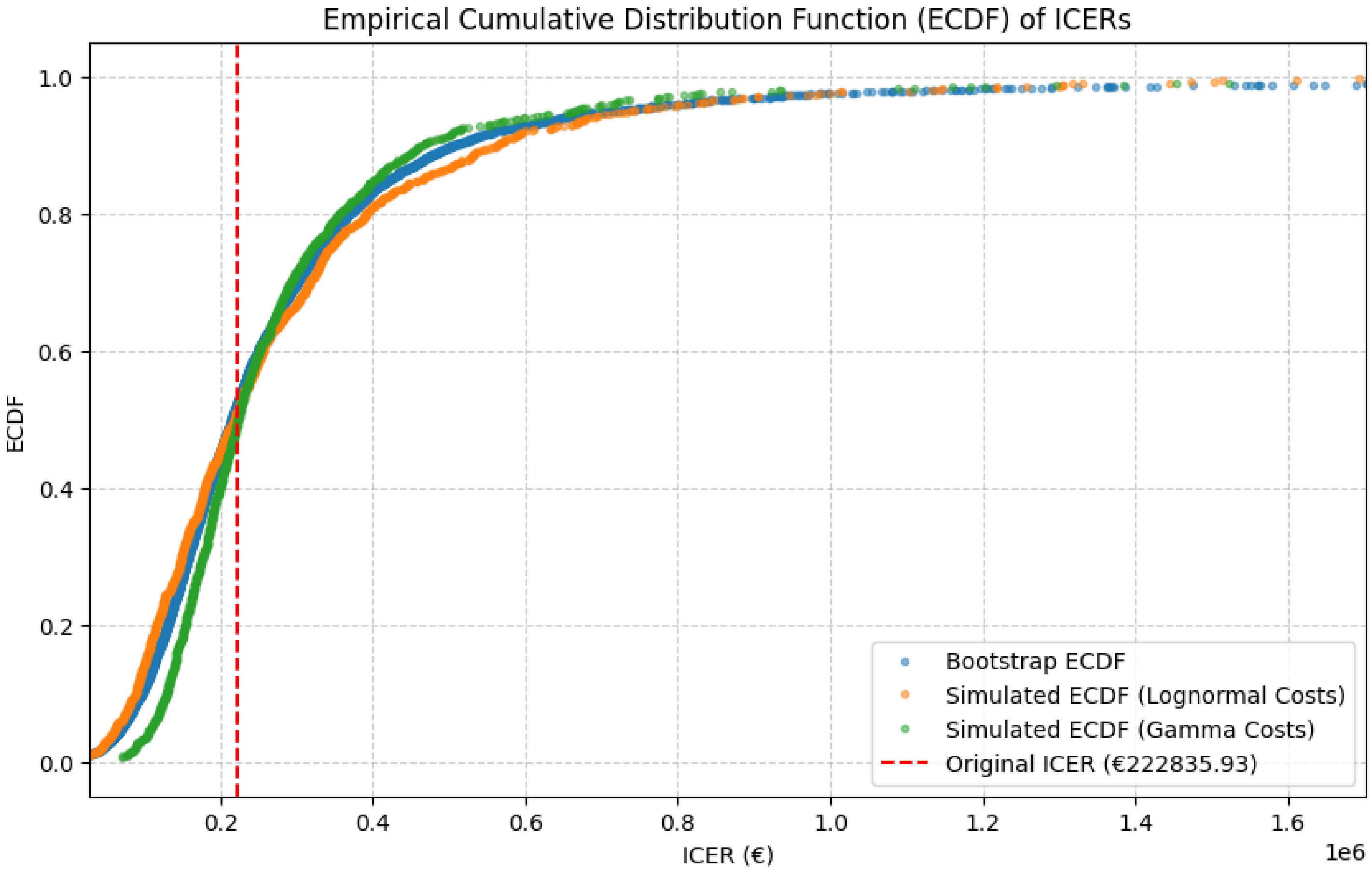

```
Quadrant Analysis of Simulated Estimates:
Quadrant 1 (More costly, More effective): 991 (99.1%)
Quadrant 2 (Less costly, More effective - Dominant): 0 (0.0%)
Quadrant 3 (Less costly, Less effective): 0 (0.0%)
Quadrant 4 (More costly, Less effective - Dominated): 9 (0.9%)
On axes: 0 (0.0%)
```

5. Comparison Bootstrap & Paramatric simulation

In the plot on the right we can see the distribution for each ICER value per method. The 3 lines intersect close to the vertical Delta-method ICER line, meaning the estimates closely agree on the same ICER mean of €222,835,93.

It suggests that the Delta ICER approximation is valid and that the parametric cost models do not distort the overall point estimate.

The main differences between the approaches lie in the tails of the distributions, reflecting differences in how each method represents uncertainty rather than bias in the central estimate.



Conclusion

The delta-method produced a point estimate of €222,836 per event, but also left us with a very wide 95% CI [€ -31,011.38 - €476,683.24]. This represents substantial statistical uncertainty due to high variability in costs and event rates

The bootstrap approach gave us a similar mean of €214,421, confirming that the parametric assumptions in the delta method did not strongly distort the estimate. The bootstrap bias was only 0.35% of the standard error, suggesting the ICER estimate is unbiased and robust.

The parametric simulations based on fitted cost distributions produced higher average ICER estimates with wide but plausible uncertainty ranges.
When lognormal cost models were used, the mean ICER became €551,053 (95% CI: €47,105–€987,970), indicating strong skewness which is consistent with the weight of cost data.

When gamma models were applied, the distribution narrowed somewhat, with a mean ICER of €321,182 (95% CI: €90,158–€827,389), suggesting a slightly more stable tendency.

Conclusion

Overall, all three approaches (bootstrap, lognorm simulation and gamma simulation) consistently indicate that the new treatment is more effective but substantially more costly, with a very wide uncertainty interval that includes both potentially cost-effective and non–cost-effective regions. Meaning, the true cost could be as low as –€31,000 or as high as €476,000.

We therefore conclude that **the estimated ICER is statistically unreliable** given the current sample size and data variability.
The combination of large cost variability, especially in group 1, and relatively small incremental effect makes the ICER inherently unstable.
As a result, we cannot state with confidence whether the new treatment provides a cost-effective improvement compared to the alternative.